

Active InferAnt Stream 001.2 ~ FieldSHIFTing for the SoftComp Intelligent S

ActiveInferAntStream | FieldSHIFTing for the SoftComp Intelligent Soft Matter Workshop | 52:54

so hello everyone uh and welcome to this session of intelligent soft matter so what we try to organize is a workshop uh on very unusual workshop and this and the structure of this Workshop is also very topic is unusual and the structure is unusual uh so this is the website what you see on this screen and uh this is about intelligence of matter and it is not uh it is emerging field so it is not yet well developed and it will be maybe in future one of the direction of the soft matter and since not much it was done in this field what we are planning in this field and what we see what can be done from different perspectives so we have people working with uh materials and soft matter and experimental right and we have people from sensors and nanop particles and how to to um to sense the environment and we have also people from decision making artificial intelligence uh which also a different field so there is a mixture of different profiles and uh today what we wanted to do because the topic is very unusual and new uh here where is um uh a description and uh and also we have a program and all and we have also profiles of the people so today what we wanted to do is uh is to in this meeting should be uh also an unusual format it will have an unusual formats that it is not only presentation of the of the talks but

also uh collaboration and initiation of
the collaboration and
discussion and uh uh we can and to to
initiate the this discussion and to make
it meaningful uh what we did is uh to we
we wanted to analyze the contributions
and also that people who will join that
they may uh know each other uh before
the meeting and also to have some ideas
of other people and also of other
directions and also other ideas of
people what they can uh bring to this
meeting and with whom they can meet and
what they already if we start writing a
paper for example we have in the plan uh
the paper

on though in those sessions we have a a
plan to write a perspective paper and
when we write this perspective paper uh
we should already prepare I and to to be
prepared and to prepare some some uh
part of this text and uh to be ready to
collaborate with people because still we
have very short time during the meeting
that's why uh I suggest so we have this
session and additionally we have a
session on grant writing session and
this is uh uh Pro will be um organized
by I leag uh and also with u uh this
will be a session traditional grant
writing and addition and and last day we
will have idea generation session which
is more open and it will be uh uh
prepared which is what we are doing now
we will will be prepared in
collaboration with uh two guests one is
uh Danielle uh fredman from California
and he is a director of the active

influence Institute and today we have another guest is the Tobias K from Netherlands and he is um uh a s of the component knowledge pixels who is behind of Nano Publications so why this is uh important so we have uh uh we have the uh in here what we we want to do is to generate some ideas before for the meeting this is the the the goal and one side we will use artificial intelligence and uh the modern tools which allowed not only to present your directions but also to get insights from uh other fields through this automatic process and this is this idea generation process through field shift and this will be presented by Daniel fredman and on other side if we have those ideas we can uh register those ideas as small pieces of information with a name and the and this can be a nanop publication which can be in a uh machine readable format and that can also allowed to build on top of this piece of knowledge so the then what we to see how we can merge all those new technologies and to see how it it is possible so first of all I would like maybe to uh that maybe our guests will tell a little bit about themself and also about this tools so first of all we start with Tobias and then with Daniel will leave maybe like this so Tobias do you want to to present your

Tool uh sure yeah

so um well I can just very quickly uh introduce myself I'm Tobias Kon I'm actually no longer based in the

Netherlands I'm still affiliated with
University there but I'm now in Zurich
Switzerland and have now we have now our
small startup called knowledge pixels
and um yeah as um Vladimir already said
we are um focusing on this concept the
technology of nanopublications
um yeah I know do you want
me to now already give a give maybe one
or two minute that's a concept of this
nanopublication and then we go further
because then we will discuss how to
include it and after this how to merge
it later but if you just uh tell what is
the concept how it works and that's will
be enough okay yes of course so as the
name says an publication is like a
publication but tiny um and on top of
that it's a publication it's not just
written in English or or any natural
language but it's uh um written in you
could say a logic language if you're a
bit familiar with semantic web or link
data Technologies it's written in the
rdf language which means uh it's made of
sentences where you have subject
predicate object and all the three of
them are globally unique
identifiers um so people around the
globe can say things and when they talk
about the same thing it will
automatically connect uh um yeah if I
may kind of um simplify it a bit um so
nanopublications are kind of these
tiny Snippets of such statements in the
simplest case it's just a single such
sentence with something is related with
that thing to that thing like I know in

the in medicine it could be this this

Gene has this

this kind of effect on this disease

often it's more than one such triple as

they are called because often statements

are a bit more complex uh than just a

simple triple and that together so these

couple of

statements then consists the main

content of nanop publication uh which is

called the assertion but then it always

has some Provenance to it meaning you can

state with another such triple where

this came from for example because you

had a study that delivered it or because

it was extracted from an existing paper

or data set or whatever and then it has

some metadata where you can say who

created it with which tool uh with a Tim

stamp with a license and these things uh

and all that together then is an nanop

publication it gets a stable persistent

identifier of its own and the the thing

that we are now working with the thing

that we providing is that you're

building a decentralized ecosystem

system for dealing with such nanop

Publications so you can push them to the

world basically to the nanop publication

Network and it can then live at

different places so when you later uh

you or somebody else want to retrieve it

again you can and because has this

formal structure you can also uh ask

very specific queries that can then

precisely be answered uh for humans to

consume but also for machines and for

applications to build up on um so we can

also on the flight create new classes
and action so we can create new
vocabulary and then talk about it and
make in Publications you have a tool for
that that's called Nano Dash that is
supposed to be user friendly so it's
still kind of Cutting Edge so we there's
still a lot of work ahead of us um but
anyway so it's kind of a low threshold
to publish these and then you and others
can query it again and um yeah we're now
in the first steps to kind of apply this
more broadly um um across different
domains and with connections to journal
and yeah there's a lot I could talk about
more but I think I've already used up my
three minutes yeah so that's exactly the
purpose of this uh uh tool how to use
this tool so it is machine readable
format and this is very short and easy
to publish and this goes in the
direction of the the new new trends of
this
uh language models then they come and
the the process of
Publications uh the dissertation right
writing projects writing and those
public Large review writing that becomes
Obsolete and already doesn't have much
sense to to have all this uh uh all all
this traditional Legacy publishing
system so instead of this it it goes in
the direction of very uh specific and
personalized uh tools and also so in the
artificial intelligence now becomes more
and more in the it goes in the more and
more in the direction of uh agent based
so instead of General tools for

everything so they have you will have a
a agent which will be uh doing some uh
work or task very personalized for you
and for for other people so you will
communicate through the agent and this
is how we see how this Publications can
enter into this domain and this is how
we connect with Daniel from active
influence Institute because the whole
Institute is focusing on the development
of
those artificial intelligent tools for
that will be uh making those agents
which will be making interaction between
humans and machine and this is one of
the examples how we can apply these two
uh uh intelligence uh soft matter to
this workshop and we can just show how
at this again this is a very early stage
but how this can be approached maybe now
Daniel you can come up and show this
tool and explain any about works
great I'm going to share my screen
okay
okay all
right
well thank you Vladimir for inviting me
and it's been really fun to collaborate
on these uh directions also Tobias
that's very exciting about the Nano pubs
I think broadly uh this is going to be
the unstructured
text getting into where the structured
information of Nano pubs is going to
supercharge it but e even today I think
there's some pretty fun and exciting at
least provocative directions to explore
um it's I'm going to push this all to

the GitHub link that's in the video description and in the side chat um okay so to kind of back up and then we'll look at this on the open source GitHub last week we had an applied active inference Symposium at The Institute and then in uh those sessions we kind of analogously had participant information presenter abstracts and so on so working with Vladimir we had some discussions and we built some tools around like what can we do to add value and context before the event how can we start to think about templates and transferable tools and practices across different related or overlapping or even totally separate open science and participation initiatives so we made this repo called Symposium so Vladimir made a bunch of uh edits and and commits to the this uh before last week and recently including adding in some open source language model methods what I'm going to show is just a few more researcher side process accelerations perhaps using perplexity that's available through the browser at perplexity Ai and here is using the the API so what I did here is I took the perplexity folder that I had applied to the in active inference Symposium participants and then adapted this in this folder called ISM stream for intelligence soft matter so that's where everything that we're going to talk about today is in here so this is it just as the folder named it just

meant different pieces for people to pull out um and explore and then also I'll note that this is all done in cursor especially as there was all already a folder of this working for a different structure but analogous tasks getting to this ISM stream content was was like 10 to 20 total prompts and cursor updates um okay so in the inputs folder Vladimir provided the information on the participants and also there's a cat so here's intelligence soft matter and then there's speakers and participants and they each have this open Alex and this works so the works are some citations and bibliography and the topics are are topics of the person okay so the what the first uh script does is it researches the presenter so everything goes into the outputs folder here's the presenter uh folder so in ad Jason and a markdown here are internet-based researchs all experimental research tooling so it's it's hard to even estimate sometimes it's totally accurate if it's little information in a common name this isn't going to be very useful if there's a ton of information and and links provided it's super useful so this this outputs a summary on the presenters um but I I kind of wanted to explore the uh working with the participant information more so that's what most of the following scripts are

so the second script two and all these are run with just P from the ism stream folder just Python 3 and then the name of the script so for the participants started off with also writing uh research report I don't know if we can look up someone who's here now to see see how accurate it is but yeah yeah g is here and it looks quite accurate okay so again sometimes it's super accurate sometimes double check okay so wrote this wrote This research report then this is where field shift comes into play so these are just sort of playing around with different combinatorics of of what is possible and also this is just using all by all folder iterating so it's not using a database it's it's hardly agentic in the sense that we could call any part of a computer pipeline agentic like is a database search a search agent I guess if we do think about it that way then it is but this is just using all by all iterating on these folders which keeps it plain text and and really available also for nanop pubs and GitHub and all this so what field shift does there's a folder called domain so in this folder called domain this is copying a folder out from a previous um repo and live stream called field shift that was based upon some work by Mike L slap so what domain has is it has these synthetic knowledge files so they're just specific questions examples factual statements within a

domain so just some of them are they're just meant to load up the context and just provide information on a synthetic domain but you could imagine that could be like a person's record or an organization's record or a different kind of document but here just being used just to say these are just statements and open questions and problems and this happened this year all those kinds of things for different domains so what three field shift does is it takes each each participant's background and it it um will find I didn't run this for every single participant but but we'll just run in cursor now so this takes each participant's background and then at basically uses perplexity to J toose researcher background and the synthetic domain and then asks what could they learn from the field and how could they contribute to the field so this is Daniel craft and RX and fur and then the way that again there's more sophisticated ways to do prompt um engineering but what's Happening Here is it just sending to perplexity here's the participant analyze the bidirectional knowledge exchange between name and participant here's the profile of the participant here's the knowledge domain just copy paste um it each of these take like so here let's just look at this one that just was made here's what this person could learn from RX iner these are accurate parts of

of what they could learn from that field
how could they contribute to the
domain I don't know but that that's from
into so that's that's field shift and
then that just iterates over whichever
domains are in the domain folder so it's
like here's statements about William
Blake

um that's uh three so that's the the
um field shift okay then Vladimir
provided a folder of catechisms so
catechism can mean different things but
in the project management setting it's a
set of question and answers that are
aligned with project management being a
grant officer program manager these
kinds of questions that are important to
be addressing so took this
for shifted

catechisms here this was like one prompt
from three copied three and said okay
now with the catechism in at catechism
folder write a that structure for each
of these shifted fields

so now with the Daniel craft blockchain
bir directional relationship now it's
going to write a
catechism with that

structure so it'll it'll take a few
seconds um then just while this one's
processing um the other way to use a c
ISM here is for all pairwise
participants write a catechism that
would reflect a

complimentary
project using the karma Gap I don't know
what it is but that catechism format so
we'll we'll look at

okay here's

five so here's Daniel craft X dong and
then that is just saying okay now for
these two

participants there's other ways that you
could fill up the context window and
everything but just saying here's how
you write the

document here's here's what here's what
we're looking for with the document

Styles here's the template of the
catechism here's the participant

information here's their other proposals
that we know from them here's their own
here's their field shifted um to and
from and then make a collaborative
project

okay first let's look at the just the
field

shift

um blockchain I I don't know this person
but we'll see you know they they can
see to and from

blockchain and then just last piece is
the collaborative proposal

we'll look at it in in

GitHub yeah maybe I can also show uh the
results when we we were Shifting the
profiles of the people because they are
coming from different domains uh to the
uh context of the workshop it's

intelligence soft matter and what was
the outcome uh so the were it was done
uh with three steps first step is to
generate profile of the each person and
also the questions uh maybe if I I

switch to online now let I I will I will

show the outcome of this uh result yes

Danielle

yes uh

okay uh uh this is here so this is also in in the same uh uh giup GitHub folder so if you go here so there is the inputs and inputs are basically they are taken from uh this is intelligence of matter and participants speakers and we have outcome is the the publication taken from open Alex so there is a publication uh directions research directions uh topics and uh also citations so it's it this is uh provided in in this format so we have uh this as a basis so we don't use any other information and we can do it for any researcher and after this uh the question was uh after we we have threee u s generate profile of the researcher with respect to the field of intelligence of matter then we have a researcher methods we it can extract the methods and also description of the methods from each profile and additionally we can do collaborative projects and the projects depending on the catechism what Daniel explained so what for example we have European projects they have a a certain structure if we have a a American project or some other funding then it will be completely different questions and different structure and it will be a differ so the catechism is here so this just a text file and for example we if we have European grants this is a certain structure if we have and here we we have

just another one this is katein for
karma projects and this is just the
questions what need to be addressed and
this is a structure first is a vision
and scope project statement proposal
solution collaboration knowledge
integration plan yes so we use this to
construct projects uh with and also
connections between people so then and
how it works so we
have it is in this folder synthetic and
here this is the also generated so first
of all we have all merged research
profiles of all participants in this
file so this is a um a automatically
generated file and the same we have
automatically generated description of
the intelligence soft matter and the and
this is current understanding future
Direction and changes so this is done in
this format and this is what is known in
general so this is basically uh
generated contracts but H you can also
have a look so this is also may be
interesting to see if you agree or not
and this is basically general knowledge
available in the internet about this
this field and future Direction so this
is again uh it is a rapidly evolving
with several excited future directions
and those are seven directions uh what
basically we it is discussed in the
literature so it is not coming from
anywhere and uh uh uh and we have also
open questions this is for fundamental
science for computational science uh
material design applications ethicals
and emerging questions so there are the

questions and future Innovation so this is again uh how to to because if we talk in the workshop about intelligence of matter we need to talk what do what do we consider it and this is our actual State maybe after the meeting it will be completely Rewritten we have a new ideas and they will be doing this and here is the beauty of this method so we can be based everything is based on agents and then we can re regenerate all the structure again this is not a problem because uh it takes maybe 10 minutes after this we go to participants and then again we can pick up the participant uh there is a uh first of all so we have a a research profile so the research profile that you already saw uh there is the research focus of the person then we impact analysis so this High cations focus on fundamental corness we have research Evolution from where it started and where this researcher goes and then there is a ke contributions and also future directions with respect to this field intelligence soft matter so uh where this profile can fit in in in this uh structure what we proposed and this and if we will if in ENT oft matter will be redefined this is fine then this all those profiles will be again regenerated and then uh another one is a uh the collaboration so we know this field we know where this researcher come from and then the question is uh what collaboration what is the matching contribution he may search and here what

we see that there are several suggestions to whom he may collaborate so it will be complementary and uh so that this is a totally uh automatic uh and there you can also have a look and then and after this uh it a third uh part so we have an second one is uh the pro we have methods so we can ex extract methods from also from Publications and then uh analyzing those publication we can see that there is a list of methods with a description with Monto simulations and then with the description and uh the result outcomes then we have also several models so this is we see all those uh theoretical models and uh we see also emerging Trends how this field evolves and limitations for example if you use scaling models scaling Theory rization group what will will be the limitations and uh this is very useful when we talk about collaboration and projects after this uh we go to Project generation and for example this is the project generated for this particular researcher and here what we see is the emerging computation and con biohybrid materials networks so here we see all description so then then uh starting from the title uh they have project rational and objectives so this we have according to this catm uh what we set up for example if we want to structure it in the in the form of European project there is a special uh setup for European project if it is in this was done for karma Gap and here we

see uh the um uh the structure for this particular project and here what we see all those topics like alignment with you you calls project Vision objectives and contribution to PO priorities expected kpis then they have St of of the art gaps and limitations novelty um and then we we see also alignment with Europeans this is uh by the way this is a European structure method methodology what can be done in this methodology expected impact and you added value and the long-term risk management and this all this resources and work plan and deliverables so this is already uh we have this dissemination exploitation so ethics and all these things so this is a European structure so we we have another project uh which was done uh here so this is another project is a which which is called a different type of project with a completely different structure and this is simplified this is karma Gap projects are simplified and here we have a vision and scope we don't have any Europe Dimension gender balance and all this things which will be uh very particularity of the European project here it is a vision and scope why this project needed now a societal need problem statement and uh proposed solution and K Innovation and what is also important that here we also see uh with this project which is another type of project with neuromorph soft met by inspired mive Network ofed nanop

particles it just suggests the person
from the
workshop who will be the best contact
with respect to this uh uh project for
example here what is this idea so we
have a neuromorphic integration so this
is what is this so this is M resistors
and how to make them from soft matter
materials using photocatalytic nanop
particles embedded in polymer Matrix so
this is the topic and uh uh it was
generated so we have a connection with
other participant and uh we have a
structure already already for the grand
proposal and it was generated in 100
seconds so it's basically two 2 minutes
and you have a project like this then
with a full details and here also what
is also included uh Network and
collaboration so who else from the
meeting can contribute to this idea and
here because we know all the profiles so
they just list those people from the
from the workshop uh and how these
people can contribute to this particular
project and uh again we have this uh uh
methods so this is again what methods
need to be taken into account they have
this St

Sam spectroscopy so computational data
biological data machine learning so then
it will be uh also the structure of the
implementation plan so this is structure
of the project we it can
be a several year project and how to
structure it and uh we have a
methodology what what else is necessary
and what is missing and you see a the it

says also that not all people from the workshop is enough that we will be again uh some extra people this uh material scientist engineer it's missing and we don't have it in the workshop but if we just limit people from the workshop it just give you this uh those names which are more suitable for this project so this is uh the the the way to operate so this is this will be just an example of course we can delete this all and it may not uh maybe not uh it will be just an example so this we can use it as an example of future collaboration so for us it is important that if you can just log in and have a look of researcher profiles it is also possible to generate for uh even for each person to whom to talk and to to which subject for example you can with your profile then you can talk to another person and also present something that maybe you don't even realize that you can work together on something on this and that's maybe we can have a list of people and to topics that you can address those people because they are experts in this field so this is more or less what we are doing now and this is how the the this uh field is developing so and I I know that Daniel want to show something else thank you Vladimir that's awesome that's cool okay I'm going to show the last piece of the script and I mean I hope this is giving some fun ways in each symposia that you all or anyone organizes or or we organize we can keep

on iterating and making it more
interesting okay let me just
share take a second

[Music]

do you

see okay let me try screen sharing
again I'll just show this for um a
minute than than any any thoughts
or questions on
this okay H says I can't share my
screen I'll try one more

time Roger

Sherin

[Music]

okay so meanwhile if you have any
questions you can

ask let me just reload I'll be back in
one second

interaction and and this is why we we
also see a big uh

um very

big potential in nanop Publications

because uh this is the way to structure
the pieces of information which is uh
relevant and which you can say this
because if you build a project what you
everything can be generated but what is
important is what is really uh
verifiable uh data and what is a
statement which is

a is a you can trust and this is where
this nanop publication come because you
don't need the whole paper you don't
need to write to read reviews but what
you need is to to have those pieces that
you can construct your own project you
can construct your own paper or
something like this and this is uh can

be also

done

automatically yes and this is where we see this interaction between AI humans and the this system of the blockchain

which is basically uh what the to

shows okay um it's it's uh it's a funny thing but just navigate over to the is M

stream and I'll show I'll show it on the

stream but just for those in the in the

chat um in the ism stream there's a

stream there's a folder six called

visualize um and what what visualize

does again it'll be visible on the

uh

stream I'll run it in Koda

2 okay first thing it does is it

iterates over each kind of the output

document so the researcher profiles I I

know you don't see it in the Google chat

but but it's visible on the on the

stream but just it just a funny thing is

happening but then um it iterates over

each kind of document like the research

profiles and the other and then it uses

standard set of linguistic analysis

methods like like topic modeling and

principal component analysis all these

will be in the in the um visualizations

folder and and so it makes for example

principal component plots and and what

those principal components um are loaded

on for different researchers and looks

at different topic modeling and

statistical aspects of of the semantics

of the event so that opens up a lot of

really interesting possibilities for

like designing unlikely connections and

setting up projects that are in certain regions and especially combined with the methods that Vladimir just mentioned Nano pubs I mean having a nano CV with available methods this is really changing the game because of the semantics being openly accessible and the rate limiting step to a large extent being like what projects are really going to happen with who and how uh yes I we see uh the ail first M MH oh okay thanks thanks a lot for this is I got the the general pictures very clear and what I don't have clear is so how can we refine the the results because all these results were obtained by automatic information that you know we're in the web but uh how do you what is the how do you iterate in such a way that you know this can be refined because this could be all junk right in PR yeah yeah yeah of course yes and this is the way that this is just to give an idea of before the meeting for of other participants and then to see what can be done and then uh to uh refine it is basically on the the field what you want to to achieve then uh the structure of of the questions what you need to to answer and uh also the profile method profile so this the basic information then it because what it does is just the shifting so example if you have a method in in polymer physics so and you want to apply it in some maybe some elements of [Music] quantum quantum theory into polymer physics so and you know that this is how

Deen got to Nobel Prize and then and and
he just go from one field to another
field and bring uh the tools within so
this is the how it does so this is just
merging different approaches which is
already common in One Direction in one
field and applying to another field and
then uh the Fe this is done with the
prompts this is prompts and there are
maybe 20 prompts which to generate uh
one document but you see that um
adjusting it then you can uh uh improve
a lot and put more details it will put a
lot so this is the short answer there
it's in GitHub so get people to have
editor access on the repo for the event
and then have it just as a starting
point and make edits in collaborative
fashion and get just like anything other
else okay thanks thanks a
lot and

rudol yeah thank you very much for this
for this very interesting presentation I
think that could be sort of a game
changer I've I I I will be very open I
do not yet really see how that act
actually works in practice but I I like
i' love to try it out man I'm a little
bit uh most of the questions have
already be answered because I had more
or less uh my questions headed into the
same directions as ailles uh but I
haven't got and that may be my fault I
apologize because I had to go to the
door briefly but how do we
operationalize this during our meeting
M right then we have to we have to
prepare something and we have to okay

and this is why I I prepared uh the coda
uh link so this is a link where you
already some people already joined uh
the join publication because we have um
publication so it is already assign
topics because they are writing what
they want to contribute and more
important that you come already with the
idea what other people are doing what is
their contribution and you can already
have ideas
of what do you want to write in and what
do you want to say about this field or
maybe this and then H is the way uh to
to to do it because then here it just
it's like a Google Document and then you
can modify it and whatever you want you
just put it there text and after this we
will discuss already this draft because
draft is there we have also the
definition of the field is also there
and it is coming from me it's it's may
be wrong so that's say but it can be
also uh then you can modify it and AI
also you can change this direction
because what we will also ask a question
what is the
minimal uh uh version of those a AI uh
all this uh intelligence in in the uh in
the materials on what are the limitation
of of experimental limitations so this
kind of things I invite you to join C if
necessary we can open more documents
there and then you can
contribute okay very good I do so no
thanks a lot yeah that's amazing man I
to me it sounds as an experiment and I
apologize man see I I'm an old guy so uh

learning new technologies is always hard
for me but let's try it will'll be
fun Vil

yes uh hello um I have a quick question
um these profiles that are automatically
generated for all us can we modify them
if we think that something is not yeah
of

course a text file you can modify it's
it will be better because if you modify
your profile that will be much better
because and and and and and we can and
and we can access them in the website
somewhere or yeah yeah yeah and it it
can be done like it's in a

open it can be um done with access yes
okay thank you yeah just to share like a
funny thing from the previous Symposium
a presenter with a name that was shared
with another researcher the perplexity
internet search on profile was 100%
incorrect so then this person looked at
it sent in an updated manually written
version but that's a shortterm type
thing as we're just experimenting on
these early iterations planning more
weeks and months out and then at a
deeper time scale having more updated
information and Clarity around what's
open and what's shared with organizers
and organizations and so on about who's
available for what and what's updated
information on their Nano Publications
and on their preferences on their
capacity the more and the more relevant
and accurate that information is enables
this kind of decentralized knowledge
collaboration using synthetic methods

better so wherever accurate information
can be shared it's like it'll be
leveraged if the information is not in
the research profile then it's not going
to come into play in the later catechism
based grant

yeah so uh we have

now few minutes before the the end and

yes

V just another quick one uh okay so what
is the best time

frame uh

until we can update our profiles so the
engine will have enough time

to uh Pro

until that's until the meeting we have
roughly 8 nine days know this this you
can modify this is this is just a a toy
model we'll see how it works and after
this we will see during the meeting
maybe for sure we will have much better
ideas that

now

okay yeah that sounds really exciting to
have months in advance clear streamlined
non overloading

Communications have accurate information
connect people and then enjoy the
possibilities for having connections at
events and symposia given that they're
only several hours long and use the
before and the during and the after to
to recognize and build on all of this
amazing work and people who are at
different career stages there's and so
many connections with how prior
publishing methods
existing grant opportunities new

publishing methods like nanopub new
grant

opportunities yeah this will will
explore a lot because European grants is
not the only option but now it is
becomes more and more the decentralized
funding that comes from um also from
this blockchain and uh decentralized
science is the same so you don't need or
even University to be researcher and
then you can collaborate through the
such tools and this is how also Daniel
operate this institute is a
multidisciplinary and distributed this
is uh the future I think

yes okay so if no more questions then uh
uh we meet each other very soon almost
in a week and uh we we will start with
with this discussion in in in life

yes okay thanks thanks a lot Vladimir
thank you thank you for your
time thank you very much thank you